

Sally Crypto Bot - trading engine v2

Summary of the revamp on branch **engine-v2** | 2026-09-27 | paper trading only, not investment advice

1. Why the revamp

The live bot opened 12 automatic trades between 2026-09-12 and 2026-09-15: 2 wins, 10 losses, net -Rp 315k, and nothing since. A code review confirmed the causes:

- **Nothing ran unless the dashboard was open.** On Vercel the in-process timer never runs, and the referenced GitHub workflow did not exist. Trading only happened while a browser polled the API.
- **Entries were fragile.** An entry needed a one-bar EMA cross, then a retest counted in *observed* bars. A missed tick meant a missed signal.
- **Risk was not controlled.** Stops were often smaller than the 0.2% round-trip cost, every position was 30% of cash, an 8-bar max hold cut winners, and exits filled at whatever price the next tick saw.
- **The backtester tested a different strategy** than the one trading live.

2. What was built

Part	What changed
Security	.env removed from git tracking. State-changing routes (settings, manual trades, backtests, Telegram test, notifications) require an ADMIN_TOKEN; the dashboard asks for it once.
Always-on engine	GitHub Actions calls POST /api/engine/tick every 5 minutes. A Postgres lease stops two ticks from trading at once. Every tick is logged; GET /api/engine/status and a dashboard indicator show when the engine last ran. Opening the dashboard no longer triggers trading.
Strategy engine v2	New src/engine/ modules: setups, position management, risk-based sizing (0.75% of equity per trade), risk brakes (daily loss, losing streak, drawdown, cooldown) and one core loop used by both live trading and backtests.
Late ticks	The engine replays every missed candle in order. Stops fill at the stop price on the candle that touched them. Entries are never back-filled at old prices.
Tests and tools	18 unit tests (node --test), including a check that stepping tick by tick gives exactly the same trades as one continuous pass. New walk-forward backtester: npm run backtest.
Bug fix	Indicators were rounded to 4 decimals, which destroyed ATR and EMA precision on coins priced under ~\$0.10. They now keep 10 significant digits.

3. How entries work now

There is no remembered "armed" event any more. On every closed candle the engine re-checks the candle history against a set of conditions. If all of them hold on the latest closed candle, it buys. By default it trades 4h candles (swing) and daily candles (trend); 15m scalping is off because it lost money in every backtest.

#	Condition	Data it looks at
1	Higher-timeframe trend up: daily close > EMA50, EMA20 > EMA50, EMA50 rising	Long history (EMA50 = 50+ candles; ~400 loaded)
2	BTC healthy: BTC daily close above its EMA50	Long history
3	Tradeable: about \$20M+ 24h volume, ATR at least 0.2% of price	Last 24h of candles
4	Uptrend on the trading timeframe: EMA20 > EMA50 and close > EMA50	Long history

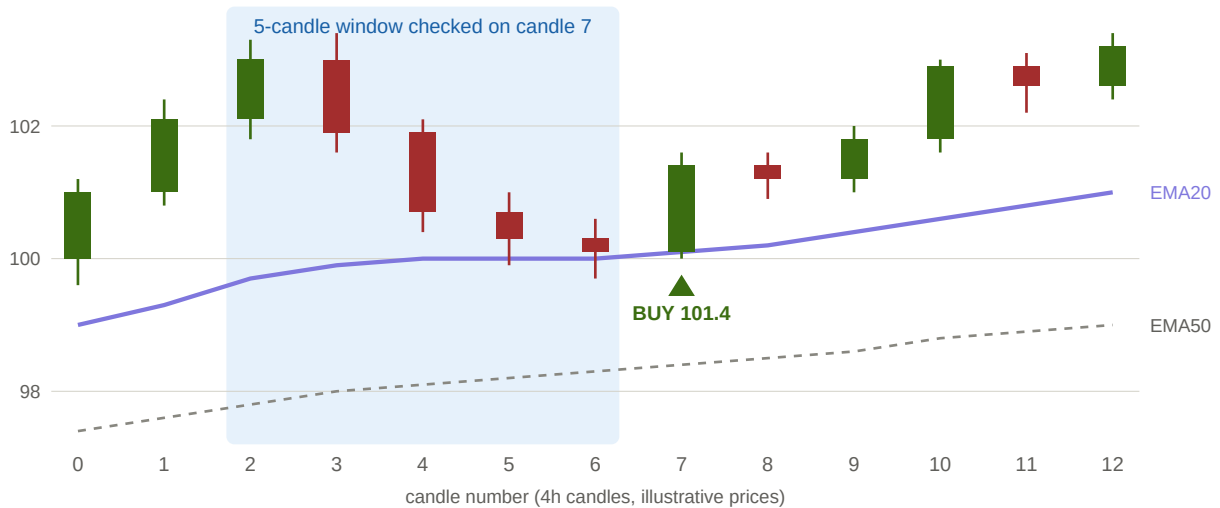
#	Condition	Data it looks at
5	Pullback happened: a low reached EMA20 (within 0.25 ATR) and RSI14 dipped to 35-55	The last 5 candles (sliding window)
6	Reclaim: close above the previous candle's high, green candle, RSI rising	Latest candle vs the one before
7	Not chasing: close at most 1 ATR above EMA20, RSI at most 70	Latest candle only

When all seven pass: the stop goes just below the pullback low (at least 1 ATR and 3x costs away; skipped if more than 3 ATR), size is set so a stop-out loses 0.75% of equity (capped at 30% of equity per position), portfolio limits are checked (max 4 positions, max 3% total open risk, no entries during a risk pause), and the fill is the close plus 0.05% slippage. Exits: breakeven once +1R is reached, then a wide ATR trailing stop so winners can run.

4. Simulation: the 5-candle window

Illustrative 4h candles (made-up prices, real v2 rules; trend and BTC conditions assumed to pass). The shaded band is the 5-candle pullback window checked on the candle that bought. It slides forward one candle at a time.

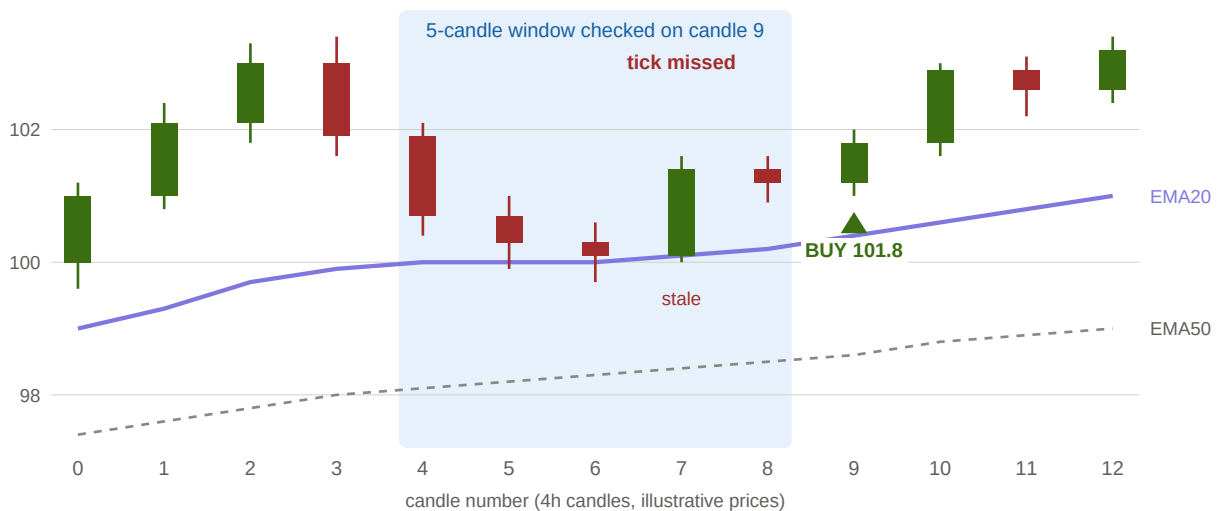
Scenario A - the engine runs on every candle



Candles 5 and 6 fail the reclaim check. On candle 7 every check passes, so the engine buys at 101.4.

Candle	Pullback in window	Reclaim	Not extended	Engine decision
5	fail (touched: none; min RSI 51)	fail (close 100.3 vs prior high 102.1)	pass (0.20 ATR)	Wait for next candle
6	pass (touched: 5; min RSI 47)	fail (close 100.1 vs prior high 101.0)	pass (0.07 ATR)	Wait for next candle
7	pass (touched: 5, 6; min RSI 45)	pass (close 101.4 vs prior high 100.6)	pass (0.87 ATR)	BUY at 101.4
8	-	-	-	Holding the position; no new entry for this coin
9	-	-	-	Holding the position; no new entry for this coin
10	-	-	-	Holding the position; no new entry for this coin
11	-	-	-	Holding the position; no new entry for this coin
12	-	-	-	Holding the position; no new entry for this coin

Scenario B - the engine misses candle 7



The engine does not run on candle 7. When it runs after candle 8 it sees candle 7 would have qualified, but does not buy at that old price. On candle 9 the pullback (candles 5-7) is still inside the 5-candle window and the reclaim passes again, so it buys at 101.8, a little later instead of never.

Candle	Pullback in window	Reclaim	Not extended	Engine decision
5	fail (touched: none; min RSI 51)	fail (close 100.3 vs prior high 102.1)	pass (0.20 ATR)	Wait for next candle
6	pass (touched: 5; min RSI 47)	fail (close 100.1 vs prior high 101.0)	pass (0.07 ATR)	Wait for next candle
7	-	-	-	Engine did not run (tick missed)
8	pass (touched: 5, 6, 7; min RSI 45)	fail (close 101.2 vs prior high 101.6)	pass (0.67 ATR)	Catch-up: candle 7 qualified but is no longer the latest candle, so it is logged as <i>missed-stale</i> (not bought at old price 101.4). Wait for next candle
9	pass (touched: 5, 6, 7; min RSI 45)	pass (close 101.8 vs prior high 101.6)	pass (0.93 ATR)	BUY at 101.8
10	-	-	-	Holding the position; no new entry for this coin
11	-	-	-	Holding the position; no new entry for this coin
12	-	-	-	Holding the position; no new entry for this coin

Old engine, same missed tick: it only compared the latest two candles for an EMA cross. The cross on candle 7 is never seen, nothing gets armed, and there is no trade until the next cross, which can take days.

5. Backtest results

15 liquid coins, 0.2% round-trip cost plus 0.05% slippage per fill, Rp 10M start. v2 settings were chosen on the older window; the recent window is the validation.

Window	Old engine (made the 12 live trades)	v2 final
Oct 2024 - Sep 2025 (tuning window)	-30.5%, max drawdown 32%	+40.6%, max drawdown 18% (optimistic: tuned here)
Oct 2025 - Sep 2026 (validation)	-5.3%, max drawdown 14%	+1.4%, max drawdown 10%

Key findings

- **Validation is roughly flat.** Closed trades averaged -0.18R; including the 4 positions still open at the end (ETH +7.5R, among others) the average is +0.06R. The validation year contained a crash (BTC -38%, most altcoins -50% to -80%) followed by a partial recovery.
- **v2 does not meet the pass criteria** set in the plan (profit factor at least 1.3 and positive expectancy after costs), so it has **not** been wired into live trading.
- **Scalping does not work at these costs.** On 15m candles, fees plus slippage were about a third of the stop distance (0.36R per trade). Every scalping variant lost money, including 1h candles.
- **Early profit-taking hurt.** Selling half at +1R removed the big winners trend-following depends on; v2 lets winners run with a wide trailing stop.
- **The trend-break exit hurt:** 443 trades averaging -0.62R with no winners; it is off in v2.

6. Status and next steps

- All work is on branch **engine-v2**; nothing is pushed or deployed. The live app still runs the old strategy.
- The Postgres lease and tick endpoint have not yet run against a real database (Docker was not running during development).
- **Decision needed:** (1) ship security + always-on engine + v2 as the paper strategy and judge it on 2 weeks of live paper trading (recommended); (2) ship security + always-on engine with the old strategy; or (3) keep researching first.

Setup required before deploying

- Rotate the Postgres password: the repository is public and .env was committed, so treat every value ever in it as leaked.
- Set ADMIN_TOKEN and ENGINE_TICK_SECRET in Vercel (generate each with: openssl rand -hex 32).
- In GitHub repository settings: add the variable ENGINE_URL (the Vercel app URL) and the secret ENGINE_TICK_SECRET.
- Scheduled workflows only run from the default branch, so the tick starts after engine-v2 is merged into main.